Rough

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Multivariable Calculus (MT1008) | Final Exam-Sp-26

Date: 02nd June, 2026 | Total Time: 3 Hours

Course Instructor(s) | Total Marks: 120

Total Questions: 13

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Important Instruction's: 1.Only simple calculator is allowed. Use of programmable calculator or any helpful devices will result F grade in the course.

2. Bonus Question (Q#14) is optional. Bonus-3 must be returned to the invigilator along with answer sheet (if attempted).

CLO #1: Defining Functions of Several Variables, Computing Partial Derivatives, Directional derivative and gradient vectors, Tangent planes and Extreme value(s). **[Weightage = 8].**

Q#1 [5]: If a differentiable function f(x, y) has a constant value c along a smooth curve

r (t) = g(t) i + h (t) j, (making the curve part of a level curve of f(x, y)), then f (g(t), h(t)) = c, show that the gradient of f(x, y) is normal to the tangent vector dr/dt, and so is normal to the curve. *Hint: Use Tree Diagram.*

OR

Q#1 [5]: Define equation for the Tangent plane and Normal line to the surface f(x, y, z) = c at $P_0(x_0, y_0, z_0)$ and use equation for the tangent plane to surface f(x, y, z) = c at $P_0(x_0, y_0, z_0)$ to deduce equation for tangent plane to the surface z = f(x, y) at $P_0(x_0, y_0, f(x_0, y_0))$.

Q#2 [5]: The surfaces $f(x, y, z) = x^2 + y^2 - 2 = 0$ (a cylinder) and $g(x, y, z) = x + z - 4 = 0$ (a plane) meet in an ellipse, find a parametric equation~~s~~ for the line tangent to the ellipse at the point $P_0(1, 1, 3)$.

Q#3 [5+5]: The derivative of $f(x,y,z)$ at a point P is greatest in the direction of v = i + j − k. In this direction, the value of the derivative is $2\sqrt{3}$. What is gradient of f (x, y, z) at P? (Give reason for your answer) and what is the derivative of f (x, y, z) at P in the direction of i + j ?

CLO #2: Evaluation of Multiple Integrals in Different Coordinate Systems and Their Application to Work, Circulation, Flux, Green's Theorem, and Stokes' Theorem. **[Weightage: 27].**

Q#4 [5]: Convert the given polar integral into a Cartesian integral and then sketch the region of integration in Cartesian coordinates. **Do not evaluate the integral.**

$$\int_0^{\pi/4}\int_0^{2\sec\theta} r^5 \sin^2\theta \, dr\, d\theta$$

Q#5 [5+5]: Find the volume of the region enclosed by the surfaces $z = x^2 + y^2$ and

$$z = \frac{(x^2+y^2+1)}{2}.$$

OR

Q#5 [5+5]: Convert the following triple integral into Spherical coordinates and then evaluate the new integral.

$$\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{\sqrt{x^2+y^2}}^{1} dzdydx.$$

Q#6 [5+5]: Let D be the region in xyz-space defined by the inequalities

$$1 \le x \le 2, \qquad 0 \le xy \le 2, \qquad 0 \le z \le 1.$$

Evaluate

$$\iiint_D (x^2y + 3xyz)\, dx\, dy\, dz,$$

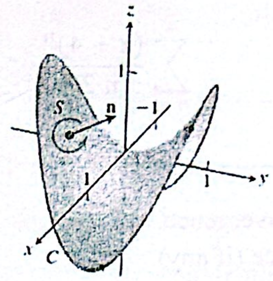
by applying the transformation $u = x,\ v = xy,\ w = 3z$, and integrating over an appropriate region G in uvw-space. Comment on the role of Jacobian.

Q#7 [1+2+7]: State **ONLY** Fundamental Theorem of Line Integral. Show that the differential form in the integral below is exact. Also, find a potential function for the field and evaluate the integral.

$$\int_{(1,2,1)}^{(2,1,1)} (2x\ln y - yz)\, dx + \left(\frac{x^2}{y} - xz\right) dy + (-xy)\, dz.$$

Q#08 [1+3+6]: State **ONLY** Green's Theorem (Circulation-Curl or Tangential form). Verify both sides of the Green's Theorem to find counterclockwise circulation of the field $F = xy\,i + y^2\,j$ around the boundary of the region enclosed by the curve $y = x^2$ and $y = x$, in the first quadrant.

Q#09 [1+3+6]: State Stoke's Theorem. Use the parameterization given below for the surface S formed by the part of the hyperbolic paraboloid $z = y^2 - x^2$ lying inside the cylinder of radius 1 around the z-axis and for the boundary curve C of S(see figure below). Then verify both sides of the Stokes' Theorem for S using the normal having positive k-component and the vector field $\boldsymbol{F} = \boldsymbol{y}\boldsymbol{i} - \boldsymbol{x}\boldsymbol{j} + \boldsymbol{x}^2\boldsymbol{k}$.



$$\varphi(r,\theta) = (r\cos\theta)i + (r\sin\theta)j + r^2(\sin^2\theta - \cos^2\theta)k, \text{ where}$$

$$0 \le r \le 1,\ 0 \le \theta \le 2\pi.$$

Hint: For L.H.S of Stokes theorem you can deduce parameterization from $\varphi(r,\theta)$.

Q#10 [1+3+6]: State Divergence Theorem. Calculate the flux of the vector field

$$F = x^2\,i + 4xyz\,j + ze^x\,k$$

out of the box-shaped region D: $0 \le x \le 3,\quad 0 \le y \le 2,\quad 0 \le z \le 1.$

Also integrate div **F** over this region and show that the result is the same value as asserted by the Divergence Theorem.

CLO #3: Some Integral Transforms, Orthogonal Functions and their applications in expanding functions into Fourier Series, Solution of IVP's using Laplace Transforms and Concepts related to convergence of Sequences, Series and Power Series. [Weightage: 15]

Q#11(a) [1+1+2]: Define Laplace Transform and use it to prove that $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$, also find $\mathcal{L}\{e^{at}t^n\}$?

Q#11(b) [06]: Solve the Initial Value Problem using Laplace transform. Also verify the solution.

$$y'' - 4y' = 6e^{3t} - 3e^{-t}, y(0) = 1, y'(0) = -1$$

Q#12 [3+7]: Expand the given function into a Fourier series.

$$f(x) = \begin{cases} x - 1, & -\pi < x < 0 \\ x + 1, & 0 \le x < \pi \end{cases}$$

Q#13 [7+3+3+2]: For the power series given below,

$$\sum_{n=1}^{\infty} \frac{(x+4)^n}{n\, 3^n}$$

investigate:

1. The interval of absolute convergence.
2. The Conditional convergence (if any).
3. The Interval of convergence.
4. The Center and Radius of the convergence.

<u>Q#14 [Bonus Question]: Attempt only one from the following, Bonus-1, 2 must be solved on the answer sheet.</u>

Bonus-1[10]: Calculate the area of a curved smooth surface S Parameterized by

r (u , v) = f(u, v) i + g (u , v) j + h (u , v) k, where $a \le u \le b,\ c \le v \le d.$

What is the condition(s) on r (u, v) of S so that S will be smooth?

<u>OR</u>

Bonus-2[10]: Suppose that F(x, y)= M(x, y)i + N(x, y) j is a velocity vector field of a fluid flowing in the plane and that the first partial derivatives of M and N are continuous at each point of the region R. Let (x, y) be a point in R and let A be a small rectangle with one corner at (x, y) that lies entirely in R. Then calculate the circulation rate per unit area for the rectangle.

<u>OR (Next Page)</u>

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Bonus-3[10]: Each part carry equal marks. This page (5-6) must be returned to the invigilator along with the answer sheet (if attempted). Fill in the blanks.

1. If f(x, y) has continuous second partial derivatives at a critical point (a, b), f has a local maximum at (a, b) if ______________________________.

2. The area of a surface z = f(x, y) over a region R in the xy-plane is given by the formula

A = ______________________________.

3. If $F = \nabla f$, then the circulation around a closed curve is -------------.

4. If $F = \nabla \times f$, then the flux across a closed surface is --------------.

5. If the line integral $\int_C \mathrm{F} \cdot dr$ has the same value for every path C joining two points A and B in D, then the integral is said to be ______________________________ and the field Fis called a ____________________ field.

6. The circulation density of a vector field $\boldsymbol{F} = M\boldsymbol{i} + N\boldsymbol{j}$ at the point (x, y) is the scalar expression ______________________________

7. Two functions f_1 and f_2 are____________________ on an interval $[a, b]$ if $< f_1, f_2 >= \int_a^b f_1(x)\, f_2(x)\, dx = 0$

8. The fundamental period of Fourier Series is ____________.

9. A convergent series that is not absolutely convergent is ______________________________.

10. Ratio Test implies convergence when $\rho = \lim_{n\to\infty} \frac{a_n}{a_{n+1}}$ *is* ______________.

11. Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series such that $a_n \le b_n$ for all $n > N$. If $\sum_{n=1}^{\infty} a_n$ converges then $\sum_{n=1}^{\infty} b_n$ ....

12. *curl grade f* =....................... or $\nabla \times \nabla f =$

13. If $F = Mi + Nj + Pk$ is a vector field with continuous second partial derivatives, then

div (curl F) =.................... or $\nabla . (\nabla \times F) =$

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14. The circulation density of a vector field is also known as ____________________.

15. Normal form of Green's Theorem ____________________.

16. A parametrized surface $\boldsymbol{r}(u,v) = f(u,v)\boldsymbol{i} + g(u,v)\boldsymbol{j} + h(u,v)\boldsymbol{k}$ is smooth if $\boldsymbol{r}_u$ and $\boldsymbol{r}_v$ are __________ and $\boldsymbol{r}_u \times \boldsymbol{r}_v$ is __________ on the interior of the parameter domain.

17. The area of the surface $F(x,y,z) = c$ over a closed and bounded plane region R is ____________________.

18. Surface area differential dS acts as a __________ factor, similar to the Jacobian in change of variables.

19. If the line integral over any closed path is zero, the field is ____________________.

20. If the curl is zero, then the line integral around a closed loop is ____________________.

Best of Luck